CODE:

```
% Graph representation
edge(a,b,4).
edge(a,c,3).
edge(b,d,5).
edge(b,e,12).
edge(c,f,7).
edge(c,g,10).
edge(d,h,2).
edge(f,i,6).
% Heuristic values
h(a,14).
h(b,12).
h(c,11).
h(d,6).
h(e,4).
h(f,8).
h(g,5).
h(h,0).
h(i,3).
% Best First Search
best_first(Start, Goal) :-
  search([[Start]], Goal).
search([[Goal|Rest]|_], Goal) :-
  reverse([Goal|Rest], Path),
  write('Path Found: '),
  write(Path).
search([Path|Paths], Goal) :-
```

```
    extend(Path, NewPaths),
    append(Paths, NewPaths, AllPaths),
    sort_paths(AllPaths, SortedPaths),
    search(SortedPaths, Goal).
extend([Node|Rest], NewPaths) :-
    findall([Next,Node|Rest],
        (edge(Node,Next,_),
         \+ member(Next,[Node|Rest])),
        NewPaths).
sort_paths(Paths, SortedPaths) :-
    predsort(compare_heuristic, Paths, SortedPaths).
compare_heuristic(<, [N1|_], [N2|_]) :-
    h(N1,H1),
    h(N2,H2),
    H1 < H2.
compare_heuristic(>, [N1|_], [N2|_]) :-
    h(N1,H1),
    h(N2,H2),
    H1 >= H2.
```

OUTPUT:

```
40 ?- consult(['c:/Users/arunp/OneDrive/Documents/AI/EXP 27.pl']).
true.

41 ?- best_first(a,h).
Path Found: [a,b,d,h]
true
```